company/business data.

In the past, mobile device management (MDM) solutions worked well, giving IT admins the management capabilities to manage COPE devices. The usual MDM solution enables IT admins to manage and govern the complete mobile device. MDM allows IT admins to wipe all data, locate devices, apply policies and generally govern COPE devices.

The rise in BYOD opens up the need for alternate mobile management strategies as MDM isn't a fit for BYOD. Let us briefly understand the rationale behind the need for alternatives.

Along with benefits, BYOD comes with challenges. With BYOD, the chances are high that enterprise data on the user's personal device can be compromised. For example, if an employee uses a smartphone to access the data on the company network and then loses that phone, untrusted parties could retrieve any unsecured data from the phone. Such risks call for an appropriate mobile strategy to secure enterprise apps and data.

MDM does not mesh well with BYOD users as they would like to keep their privacy intact while they use their smartphones for work. Users may not like MDM solutions to completely govern their personal devices, which may have their personal files along with business apps and data. Users will be reluctant to use their devices for work if the governing MDM solution is always keeping an eye on their geo location or if there is a chance that their personal files may get accidentally wiped by the IT (MDM) admin.

On the other hand, IT admins may like to at least have basic mobile management capabilities, even for BYOD, so as to effectively do the following:

- *Distribute business applications (in-house or third party) from one place.*
- *Secure data on the move.* Company employees will run business apps on their mobile devices and may fetch business data over public/open Wi-Fi or 2G/3G networks. This confidential data getting transferred over networks while users are on the move is termed as data on the move. It becomes important to take measures to safeguard this data as it can be sniffed and compromised on open networks.
- *Secure data at rest.* Company employees may save business data/files locally on their mobile devices while using business apps. These confidential data files stored locally can be extracted by anyone who gets access to the mobile device. Thus it becomes important to safeguard these data files.
- *Protect data leaks.* There could be a few business mobile apps serving very sensitive data to the end user, like the price quotations of a product. IT admins may want to restrict the screen capture or copy/paste of such sensitive pieces of information from the application.
- *Configure the application.* IT admins may want to set up business apps for their employees by dynamically configuring key parameters like the enterprise backend server URL and port, where the application should connect or fetch data from.

Containerisation and MAM – the MDM alternate strategy best suited for BYOD

The need for the earlier mentioned mobile management capabilities and the partial mismatch between what MDM does and BYOD users want, have resulted in containerisation. This is a new methodology which separates business data from personal data on the user's device. This methodology creates a separate and secure storage space on the device to store business apps and data, away from personal data. This space can be thought of as a separate container/box which keeps business apps and data secure in silos, away from intruders.

Mobile application management (MAM) is also gaining popularity with containerisation. MAM is about managing just the business apps used for business operations instead of managing the entire device. MAM and containerisation go hand in hand, and have become the mobile management strategy for BYOD.

The older EMM integration methodologies for containerisation and MAM

Initially, EMM (enterprise mobility management) vendors devised diverse integration methodologies to achieve containerisation and MAM. These had some benefits as well as quite a few downsides.

Let me highlight a couple of old methodologies along with their pros and cons, which I experienced while doing some WFM MAM work for iOS and Android.

1. ***MAM (EMM) SDK integration methodology:*** In this methodology, the developer needs to integrate the EMM propriety mobile MAM SDK code into the mobile application code. Each EMM propriety MAM SDK library code will be different, and will provide varying mobile management feature sets. There are several EMM solutions in the market and, with this approach, MAM SDK code for each of these EMMs has to be plugged in with the mobile application to support them all. The following are the pros and cons of this approach.

Pros

- Full blown MAM feature support.
- Fine grain management and control.
- Possibility of extended/custom management and security features.

Cons

- Can only support internal mobile apps with MAM SDK. Chances of managing public mobile apps from third party ISVs are less as the app may not have EMM SDK code in it.
- EMM vendor lock-in, if the mobile application is integrated with the MAM SDK code of just one EMM. To support the new EMM, MAM SDK code of the new EMM has to be plugged in within the app code.
- Multiple EMM MAM SDK codes in a single mobile application will increase the following: